

Assignment 1

1.Data Import:

- Import List of Orders.csv into Power BI.
- Open the list of orders in the Power Query Editor using the Transform Data option.
- Import Order Details.csv and Sales Target.csv into Power Query Editor.

Queries [3]

fx = Table.TransformColumnTypes(#"Promoted Headers",{{"Month of Order Date", type date}, {"Category", type text},

	Month of Order Date	Category	Target
1	01-04-2018	Furniture	10400
2	01-05-2018	Furniture	10500
3	01-06-2018	Furniture	10600
4	01-07-2018	Furniture	10800
5	01-08-2018	Furniture	10900
6	01-09-2018	Furniture	11000
7	01-10-2018	Furniture	11100
8	01-11-2018	Furniture	11300
9	01-12-2018	Furniture	11400
10	01-01-2019	Furniture	11500
11	01-02-2019	Furniture	11600
12	01-03-2019	Furniture	11800
13	01-04-2018	Clothing	12000
14	01-05-2018	Clothing	12000
15	01-06-2018	Clothing	12000
16	01-07-2018	Clothing	14000
17	01-08-2018	Clothing	14000
18	01-09-2018	Clothing	14000
19	01-10-2018	Clothing	16000
20	01-11-2018	Clothing	16000
21	01-12-2018	Clothing	16000
22	01-01-2019	Clothing	16000
23	01-02-2019	Clothing	16000

Query Settings

PROPERTIES

Name

Sales target

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Observation:

I have successfully imported three CSV files into the Power BI Power Query Editor: **List of Orders.csv**, **Order Details.csv**, and **Sales Target.csv**. All tables are visible on the left side under the Queries pane, and the data is loaded correctly.

2.Row Limiting & Data Types:

- Restrict the List of Orders table to the first 500 rows.
- Set the Order Date column to the Date data type.
- Change the Amount and Target columns to fixed decimal numbers.

Queries [3]

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✓

f_x

= Table.TransformColumnTypes("#Reduce rows first 500",{{"Order Date", type date}})

▼

A^BC Order ID ▼

A^BC Order Date ▼

A^BC CustomerName ▼

A^BC State ▼

A^BC City ▼

1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata

Sales target

Order Details

List of Orders

Query Settings

✕

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source ✖

Promoted Headers ✖

Changed Type

Reduce rows first 500 ✖

✕ Order Date Changed Date type

Queries [3]

Sales target

Order Details

List of Orders

fx

= Table.TransformColumnTypes("#Changed Type",{{"Amount", Currency.Type}})

Order ID	\$ Amount	Profit	Quantity	Category	Sub-Category
1 B-25601	1,275.00	-1148		7 Furniture	Bookcases
2 B-25601	66.00	-12		5 Clothing	Stole
3 B-25601	8.00	-2		3 Clothing	Hankerchief
4 B-25601	80.00	-56		4 Electronics	Electronic G
5 B-25602	168.00	-111		2 Electronics	Phones
6 B-25602	424.00	-272		5 Electronics	Phones
7 B-25602	2,617.00	1151		4 Electronics	Phones
8 B-25602	561.00	212		3 Clothing	Saree
9 B-25602	119.00	-5		8 Clothing	Saree
10 B-25603	1,355.00	-60		5 Clothing	Trousers
11 B-25603	24.00	-30		1 Furniture	Chairs
12 B-25603	193.00	-166		3 Clothing	Saree
13 B-25603	180.00	5		3 Clothing	Trousers
14 B-25603	116.00	16		4 Clothing	Stole
15 B-25603	107.00	36		6 Clothing	Stole
16 B-25603	12.00	1		2 Clothing	Hankerchief
17 B-25603	38.00	18		1 Clothing	Kurti
18 B-25604	65.00	17		2 Clothing	T-shirt
19 B-25604	157.00	5		9 Clothing	Saree
20 B-25605	75.00	0		7 Clothing	Saree
21 B-25606	87.00	4		2 Clothing	Shirt
22 B-25607	50.00	15		4 Clothing	Leggings
23 B-25608	1,364.00	-1864		5 Furniture	Tables

Query Settings

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Amount changed type Fixed ...

	Month of Order Date	Category	\$ Target
1	01-04-2018	Furniture	10,400.00
2	01-05-2018	Furniture	10,500.00
3	01-06-2018	Furniture	10,600.00
4	01-07-2018	Furniture	10,800.00
5	01-08-2018	Furniture	10,900.00
6	01-09-2018	Furniture	11,000.00
7	01-10-2018	Furniture	11,100.00
8	01-11-2018	Furniture	11,300.00
9	01-12-2018	Furniture	11,400.00
10	01-01-2019	Furniture	11,500.00
11	01-02-2019	Furniture	11,600.00
12	01-03-2019	Furniture	11,800.00
13	01-04-2018	Clothing	12,000.00
14	01-05-2018	Clothing	12,000.00
15	01-06-2018	Clothing	12,000.00
16	01-07-2018	Clothing	14,000.00
17	01-08-2018	Clothing	14,000.00
18	01-09-2018	Clothing	14,000.00
19	01-10-2018	Clothing	16,000.00
20	01-11-2018	Clothing	16,000.00
21	01-12-2018	Clothing	16,000.00
22	01-01-2019	Clothing	16,000.00
23	01-02-2019	Clothing	16,000.00

PROPERTIES

Name

Sales target

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Target changed Fixed type de...

Preview based on top 1000 rows

PREVIEW DOWNLOADED AT 2

Step 1: Row Limiting

- **What we did:** Restricted the **List of Orders** table to show only the first 500 rows.
- **Tool used:** **Keep Top Rows** option under the *Home* tab (Reduce Rows section).

Step 2: Order Date Data Type

- **What we did:** Verified and set the **Order Date** column to the proper 'Date' format.
- **Tool used:** **Data Type Icon** (Change Type) next to the column header.

Step 3: Amount Data Type

- **What we did:** Changed the **Amount** column in both **List of Orders** and **Order Details** tables to decimal format.
- **Tool used:** **Fixed Decimal Number** option from the Data Type menu.

Step 4: Target Data Type

- **What we did:** Updated the **Target** column in the **Sales Target** table to decimal format.
- **Tool used:** **Fixed Decimal Number** option from the Data Type menu.

3.Text Formatting:

- Format the CustomerName column to proper case to ensure consistent capitalization.

Queries [3] | Sales target | Order Details | List of Orders

fx = Table.TransformColumns(#"Changed Type",{{"CustomerName", Text.Proper, type text}})

	A ^B _C Order ID	Order Date	A ^B _C CustomerName	A ^B _C State	A ^B _C City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata

Query Settings

PROPERTIES

Name
List of Orders

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

CustomerName first letter Ca...

Observation:

- **What we did:** Formatted the **Customer Name** column to proper case to ensure uniform and consistent capitalization across all records.
- **Tool used:** **Capitalize Each Word** option under the *Transform* tab (Text Column section).

4. Column Creation:

- Merge the City and State columns to create a new column named Location in the format: City, State
- Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

Observation:

- **Column Merging:** Merged the **City** and **State** columns using a comma separator to create a new location column named **Location**. (**Tool used:** *Merge Columns* under *Add Column* tab).
- **Custom Column:** Created a calculated column named **Profit Margin** using the formula $\frac{[\text{Profit}]}{[\text{Amount}]}$ and set its format to percentage for financial tracking. (**Tool used:** *Custom Column* under *Add Column* tab).

5. Conditional Column:

- Add a conditional column named **Profit Status** based on:
 - Profit < 0 → Loss
 - Profit = 0 → Break-Even
 - Profit > 0 → Profit

Queries [3] | Query Settings

fx = Table.AddColumn(#"Profit Margin Changed Percentage type", "Profit Status", each if [Profit] < 0 then "Loss" else if [Profit] = 0 then "Break-Even" else "Profit")

	Quantity	Category	Sub-Category	% Profit Margin	Profit Status
1	-1148	Furniture	Bookcases	-90.04%	Loss
2	-12	Clothing	Stole	-18.18%	Loss
3	-2	Clothing	Hankerchief	-25.00%	Loss
4	-56	Electronics	Electronic Games	-70.00%	Loss
5	-111	Electronics	Phones	-66.07%	Loss
6	-272	Electronics	Phones	-64.15%	Loss
7	1151	Electronics	Phones	43.98%	Profit
8	212	Clothing	Saree	37.79%	Profit
9	-5	Clothing	Saree	-4.20%	Loss
10	-60	Clothing	Trousers	-4.43%	Loss
11	-30	Furniture	Chairs	-125.00%	Loss
12	-166	Clothing	Saree	-86.01%	Loss
13	5	Clothing	Trousers	2.78%	Profit
14	16	Clothing	Stole	13.79%	Profit
15	36	Clothing	Stole	33.64%	Profit
16	1	Clothing	Hankerchief	8.33%	Profit
17	18	Clothing	Kurti	47.37%	Profit
18	17	Clothing	T-shirt	26.15%	Profit
19	5	Clothing	Saree	3.18%	Profit
20	0	Clothing	Saree	0.00%	Break-Even
21	4	Clothing	Shirt	4.60%	Profit
22	15	Electronics	Leggings	30.00%	Profit
23	-1864	Furniture	Table	-136.66%	Loss

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Profit/Amount new column Pr...
- Profit Margin Changed Perce...
- Profit Checking Conditional Co...

Observation:

- **What we did:** Created a conditional column named **Profit Status** to categorized order outcomes into 'Loss', 'Break-Even', or 'Profit' based on the value of the **Profit** column.
- **Tool used:** **Conditional Column** option under the *Add Column* tab.

6. Merging Data (Joins)

- Merge the List of Orders and Order Details tables using Order ID.
- Name the merged output table as Orders Data.

The screenshot displays the Power BI Desktop interface. On the left, the 'Queries' pane shows a list of queries: 'Sales target', 'Order Details', 'List of Orders', and 'Orders Data'. The 'Orders Data' query is selected and highlighted with a red box. The main area shows a data table with columns: 'Location', 'Order Details.Order ID', 'Order Details.Amount', 'Order Details.Profit', and 'Order Details.Quantity'. The table contains 23 rows of data. A red box highlights the 'Order Details.Order ID' column. On the right, the 'Query Settings' pane is open, showing the 'Name' field set to 'Orders Data' and the 'Source' field set to 'List of Order and Order Detail...'. The 'APPLIED STEPS' section is also visible.

	Location	Order Details.Order ID	Order Details.Amount	Order Details.Profit	Order Details.Quantity
1	Ahmedabad,Gujarat	B-25601	1275	-1148	
2	Ahmedabad,Gujarat	B-25601	66	-12	
3	Ahmedabad,Gujarat	B-25601	8	-2	
4	Ahmedabad,Gujarat	B-25601	80	-56	
5	Pune,Maharashtra	B-25602	168	-111	
6	Pune,Maharashtra	B-25602	424	-272	
7	Pune,Maharashtra	B-25602	2617	1151	
8	Pune,Maharashtra	B-25602	561	212	
9	Pune,Maharashtra	B-25602	119	-5	
10	Bhopal,Madhya Pradesh	B-25603	1355	-60	
11	Bhopal,Madhya Pradesh	B-25603	24	-30	
12	Bhopal,Madhya Pradesh	B-25603	193	-166	
13	Bhopal,Madhya Pradesh	B-25603	180	5	
14	Bhopal,Madhya Pradesh	B-25603	116	16	
15	Bhopal,Madhya Pradesh	B-25603	107	36	
16	Bhopal,Madhya Pradesh	B-25603	12	1	
17	Bhopal,Madhya Pradesh	B-25603	38	18	
18	Jaipur,Rajasthan	B-25604	65	17	
19	Jaipur,Rajasthan	B-25604	157	5	
20	Kolkata,West Bengal	B-25605	75	0	
21	Bangalore,Karnataka	B-25606	87	4	
22	Kashmir,Jammu and Kashmir	B-25607	50	15	
23	Chennai Tamil Nadu	B-25608	1364	-1864	

Observation:

Table Merging: Merged the **List of Orders** and **Order Details** tables using the common column **Order ID**.

New Table Creation: Saved the merged output as a new table and renamed it to **Orders Data** as requested.

Tool used: **Merge Queries as New** option under the **Home** tab.

7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.
- Define and apply appropriate strategies such as:
 - Replacing missing values
 - Removing records
 - Retaining nulls with justification
- Identify duplicate rows and define a suitable handling strategy based on business logic.

Queries [4] ✕ ✓ f_x = Table.Distinct("#City and State Column merged", {"Order ID"}) ▼

	A ^B Order ID	Order Date	A ^B CustomerName	A ^B State	A ^B City	A ^B Location
1	8-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,
2	8-25602	01-04-2018	Pearl	Maharashtra	Pune	Pune, Mahari
3	8-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal, Madh
4	8-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Jaipur, Rajast
5	8-25605	05-04-2018	Kasheen	West Bengal	Kolkata	Kolkata, West
6	8-25606	06-04-2018	Hazel	Karnataka	Bangalore	Bangalore, Ka
7	8-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir, Jam
8	8-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai, Tam
9	8-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow, Utt
10	8-25610	09-04-2018	Yogesh	Bihar	Patna	Patna, Bihar
11	8-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananth
12	8-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh, f
13	8-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh, f
14	8-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Simla, Himach
15	8-25615	15-04-2018	Bhavna	Sikkim	Gangtok	Gangtok, Sikk
16	8-25616	15-04-2018	Kanak	Goa	Goa	Goa, Goa
17	8-25617	17-04-2018	Sagar	Nagaland	Kohima	Kohima, Nagal
18	8-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Hyderabad, A
19	8-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ahmedabad,
20	8-25620	20-04-2018	Sarita	Maharashtra	Pune	Pune, Mahari
21	8-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Bhopal, Madh
22	8-25622	22-04-2018	Monisha	Rajasthan	Jaipur	Jaipur, Rajast
23	8-25623	22-04-2018	Atharv	West Bengal	Kolkata	Kolkata, West

Query Settings ✕

PROPERTIES

Name
List of Orders

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Reduce rows first 500
- Order Date Changed Date type
- Customer Name first letter ca...
- City and State Column merged
- ✕ Order ID Removed Duplicates

Observation:

- **Handling Missing Data:** Scanned all tables and identified that there are no missing or `null` values in the dataset. Therefore, no data replacement or record removal strategy was required for missing values.
- **Handling Duplicate Data:** Applied business logic to remove duplicate rows based on `Order ID` in the primary `List of Orders` table using the **Remove Duplicates** tool to ensure unique order logs.
- **Data Retention Logic:** Retained repeated `Order ID` rows in the `Order Details` and merged `Orders Data` tables, as they represent distinct product line items within the same transaction rather than duplicates.

8. Sorting and Filtering Data

- Sort records in the Orders Data table by Order Date in descending order.
- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Queries [4] fx = Table.Sort("#List of Orders and Order Details Merge",{"Order Date", Order.Descending})

Order ID	Order Date	CustomerName	State	City	Location
B-26100	31-03-2019	nitika	Madhya Pradesh	Indore	Indore,Madr
B-26100	31-03-2019	nitika	Madhya Pradesh	Indore	Indore,Madr
B-26100	31-03-2019	nitika	Madhya Pradesh	Indore	Indore,Madr
B-26099	30-03-2019	hishm	Maharashtra	Mumbai	Mumbai,Ma
B-26099	30-03-2019	hishm	Maharashtra	Mumbai	Mumbai,Ma
B-26099	30-03-2019	hishm	Maharashtra	Mumbai	Mumbai,Ma
B-26099	30-03-2019	hishm	Maharashtra	Mumbai	Mumbai,Ma
B-26098	29-03-2019	inky	Jammu and Kashmir	Kashmir	Kashmir,Jam
B-26098	29-03-2019	inky	Jammu and Kashmir	Kashmir	Kashmir,Jam
B-26098	29-03-2019	inky	Jammu and Kashmir	Kashmir	Kashmir,Jam
B-26098	29-03-2019	inky	Jammu and Kashmir	Kashmir	Kashmir,Jam
B-26098	29-03-2019	inky	Jammu and Kashmir	Kashmir	Kashmir,Jam
B-26098	29-03-2019	inky	Jammu and Kashmir	Kashmir	Kashmir,Jam
B-26098	29-03-2019	inky	Jammu and Kashmir	Kashmir	Kashmir,Jam
B-26096	28-03-2019	tharv	West Bengal	Kolkata	Kolkata,Wes
B-26097	28-03-2019	ini	Karnataka	Bangalore	Bangalore,Ka
B-26097	28-03-2019	ini	Karnataka	Bangalore	Bangalore,Ka
B-26096	28-03-2019	tharv	West Bengal	Kolkata	Kolkata,Wes
B-26097	28-03-2019	ini	Karnataka	Bangalore	Bangalore,Ka
B-26096	28-03-2019	tharv	West Bengal	Kolkata	Kolkata,Wes
B-26096	28-03-2019	tharv	West Bengal	Kolkata	Kolkata,Wes
B-26097	28-03-2019	ini	Karnataka	Bangalore	Bangalore,Ka
B-26097	28-03-2019	ini	Karnataka	Bangalore	Bangalore,Ka
B-26097	28-03-2019	ini	Karnataka	Bangalore	Bangalore,Ka

4 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name
Orders Data

All Properties

APPLIED STEPS

Source
List of Orders and Order Deta...
Orders Data Sort Descending

PREVIEW DOWNLOADED AT 23:

Queries [4] fx = Table.SelectRows("#Orders Data Sort Descending", each ([State] = "Tamil Nadu"))

Order ID	Order Date	CustomerName	State	City	Location
B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tam
B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tam
B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tam
B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tam
B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tam
B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tam
B-26018	14-02-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tam
B-26018	14-02-2019	Aarushi	Tamil Nadu	Chennai	Chennai,Tam
B-26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tam
B-26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tam
B-26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tam
B-26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	Chennai,Tam
B-25860	15-11-2018	Akshay	Tamil Nadu	Chennai	Chennai,Tam
B-25788	21-09-2018	Dinesh	Tamil Nadu	Chennai	Chennai,Tam
B-25716	11-07-2018	Surabhi	Tamil Nadu	Chennai	Chennai,Tam
B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tam
B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tam
B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tam
B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tam
B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tam
B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	Chennai,Tam
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tam
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tam

4 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES

Name
Orders Data

All Properties

APPLIED STEPS

Source
List of Orders and Order Deta...
Orders Data Sort Descending
State Column Filtered Rows

PREVIEW DOWNLOADED AT 23:

Observation:

- **Data Sorting:** Sorted the records in the **Orders Data** table by **Order Date** in descending order to review the most recent transactions first.
- **Data Filtering:** Applied a specific filter to the **State** column to isolate and analyze orders coming only from **Tamil Nadu**.
- **Tools used:** **Sort Descending** and **Filter Dropdown** options on the column headers in Power Query Editor.

9. Grouping and Aggregating Data

- Duplicate the Order Details table and perform the following:
 - Count of Order ID
 - Average Profit by Category
 - Total Amount by Sub-Category
- Duplicate the Sales Target table and aggregate the total Target by Month of order date.

The screenshot displays the Power Query Editor interface. On the left, the 'Queries' list shows 'Order Count Summary' selected. The main area shows a table with two columns: 'Order ID' and 'Total Orders'. The table is grouped by 'Order ID'. The formula bar at the top shows the M code: `= Table.Group(#"Profit Cheking Conditional Column", {"Order ID"}, {"Total Orders", each Table.RowCount(_),`

Order ID	Total Orders
B-25601	4
B-25602	5
B-25603	8
B-25604	2
B-25605	1
B-25606	1
B-25607	1
B-25608	4
B-25609	2
B-25610	6
B-25611	1
B-25612	1
B-25613	1
B-25614	2
B-25615	1
B-25616	4
B-25617	1
B-25618	2
B-25619	1
B-25620	1
B-25621	3
B-25622	1
B-25623	4

The right pane shows the 'Query Settings' for 'Order Count Summary'. Under 'APPLIED STEPS', the step 'Count of Order ID in Order D...' is highlighted.

Queries [8] ✕ ✓ *f_x* = Table.Group(#"Target Changed type Fixed decimal Number", {"Month of Order Date"}, {{ "Total Target", each List.Sum

	Month of Order Date	L2 Total Target
1	01-04-2018	31400
2	01-05-2018	31500
3	01-06-2018	31600
4	01-07-2018	33800
5	01-08-2018	33900
6	01-09-2018	34000
7	01-10-2018	36100
8	01-11-2018	36300
9	01-12-2018	36400
10	01-01-2019	43500
11	01-02-2019	43600
12	01-03-2019	43800

Query Settings ✕

PROPERTIES

Name
Sales Target Summary

All Properties

APPLIED STEPS

Source ✕

Promoted Headers ✕

Changed Type

Target Changed type Fixed de...

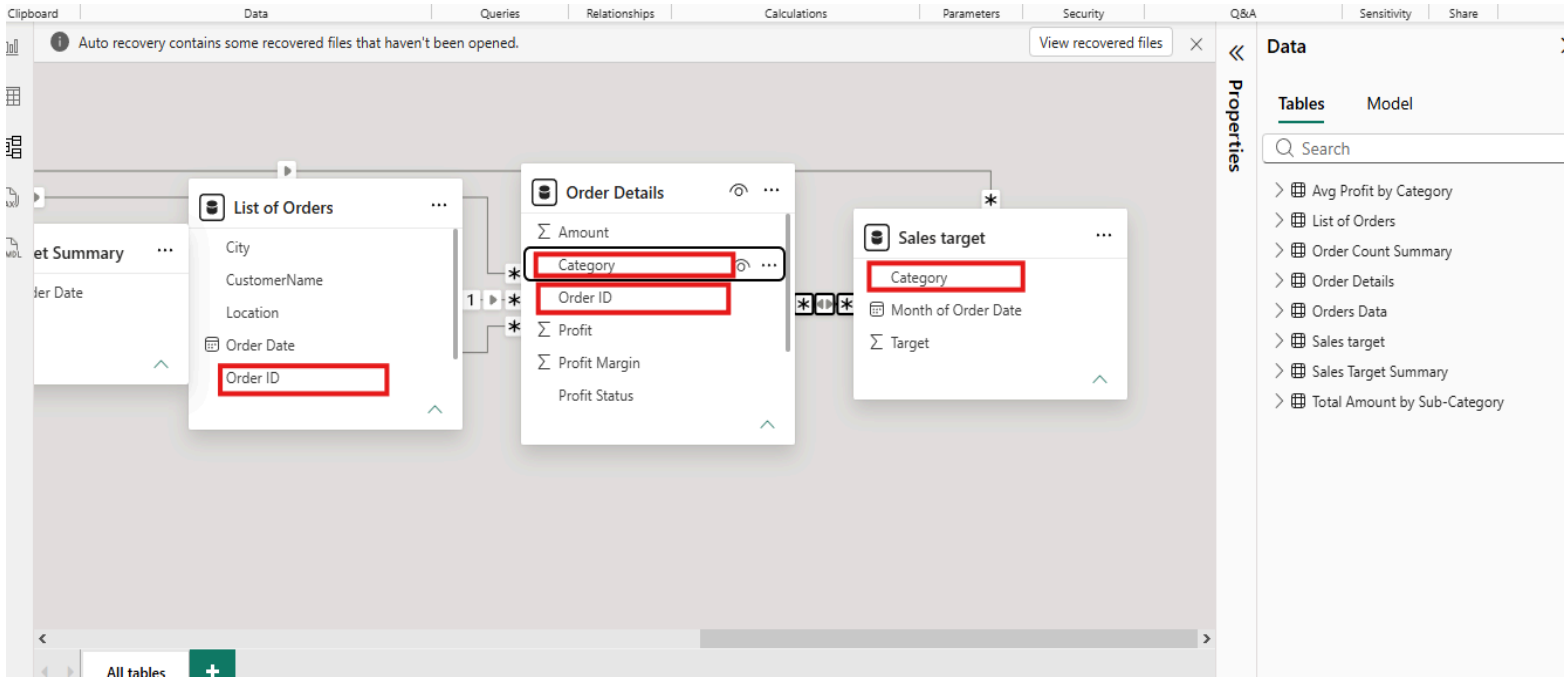
✕ Total Target by Sales target in... ✕

Observation:

- **Order Details Aggregations:** Duplicated the **Order Details** table to perform key summaries: grouped by **Order ID** to count rows, grouped by **Category** to calculate the average profit, and grouped by **Sub-Category** to find the total sales amount.
- **Sales Target Aggregations:** Duplicated the **Sales Target** table and grouped it by the **Month of order date** column to aggregate the total targeted sales amount per month.
- **Tool used:** Group By feature under the *Home* tab in Power Query Editor.

10. Data Modeling:

- **Establish a relationship between:**
 - List of Orders and Order Details using Order ID
 - **Establish a relationship between:**
 - Order Details and Sales Target using Category
- 2. Use Manage Relationships to ensure relationships are active and correctly defined.**



Manage relationships

+ New relationship Autodetect Edit Delete Filter				
☐	From: table (column) ↑	Relationship	To: table (column)	Status
☐	Order Count Summary (Order ...	1 —> 1	List of Orders (Order ID)	Active ...
☐	Order Details (Category)	* —> 1	Avg Profit by Category (Categ...	Active ...
☐	Order Details (Category)	* —> *	Sales target (Category)	Active ...
☐	Order Details (Order ID)	* —> 1	List of Orders (Order ID)	Active ...
☐	Order Details (Sub-Category)	* —> 1	Total Amount by Sub-Categor...	Active ...
☐	Orders Data (Order Details.Cat...	* —> 1	Avg Profit by Category (Categ...	Active ...
☐	Orders Data (Order Details.Su...	* —> 1	Total Amount by Sub-Categor...	Active ...
☐	Orders Data (Order ID)	* —> 1	List of Orders (Order ID)	Active ...
☐	Sales target (Category)	* —> 1	Avg Profit by Category (Categ...	Active ...

Observation:

- **First Relationship:** Established a One-to-Many (1:*) relationship between the **List of Orders** and **Order Details** tables using the common **Order ID** column.
- **Second Relationship:** Established a Many-to-Many (*:*) relationship between the **Order Details** and **Sales target** tables using the common **Category** column.
- **Verification:** Used the Manage Relationships window to ensure that both connections are correctly defined, verified, and set to Active status for seamless cross-filtering.